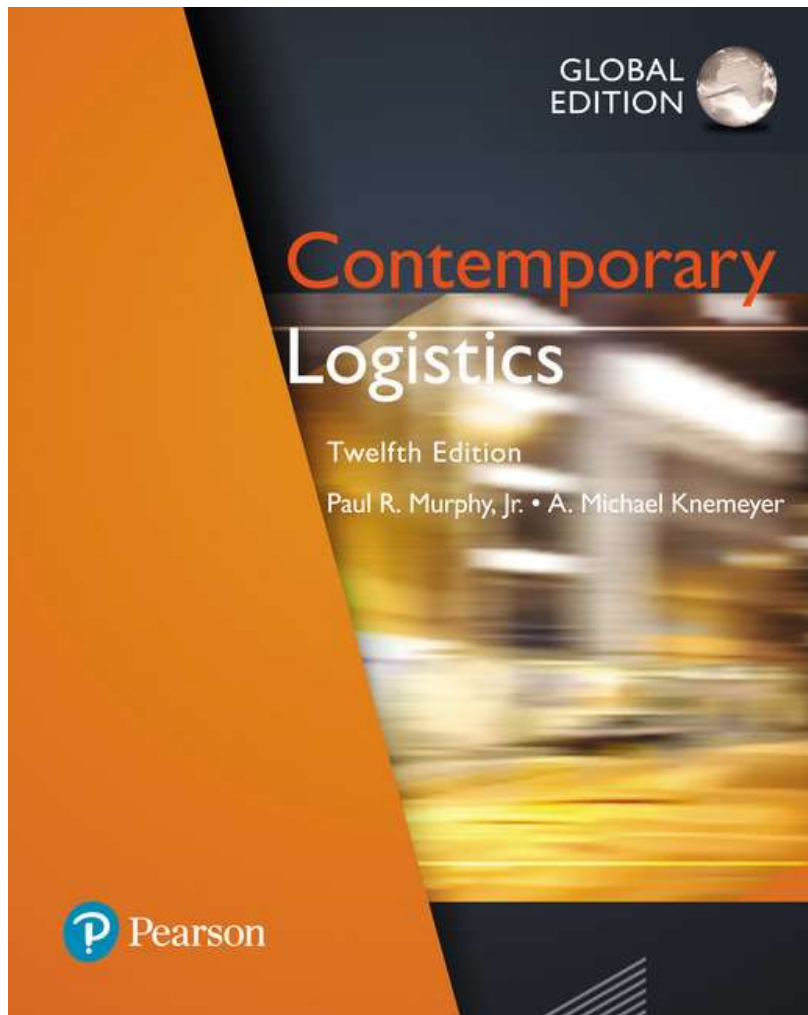


# Contemporary Logistics

Twelfth Edition, Global Edition



## Chapter 12 Transportation

# Learning Objectives (1 of 2)

- 12.1** To compare and contrast transportation infrastructures in several countries
- 12.2** To identify the five modes of transportation and learn about their respective characteristics
- 12.3** To discuss intermodal transportation
- 12.4** To describe several types of transportation specialists

# Learning Objectives (2 of 2)

- 12.5** To explain how different types of regulation impact transportation
- 12.6** To identify the legal classification of transportation carriers

# Transportation (2 of 4)

- Transportation
  - The actual, physical movement of goods and people between two points
- Transportation influences or is influenced by the following logistics activities:
  - Transportation costs are affected by the location of firm's plants, warehouses, vendors, retailers, and customers
  - Inventory requirements depend on transportation mode used (High speed \$ cost vs Low speed & cost)
  - Transport mode selected influences packaging, and carrier classification rules dictate packing choice

# Transportation (3 of 4)

- Transportation influences or is influenced by the following logistics activities:
  - Type of carrier used dictates a manufacturing plant's materials handling equipment and the design of receiving and shipping docks
  - Order-management philosophy that encourages maximum consolidation of shipments between common points enables a company to give larger shipments to its carriers and take advantage of volume discounts
  - Customer service goals influence the type and quality of carrier and carrier service selected by the seller

# Transportation (4 of 4)

- Five different types, or modes, of transportation include:
  - Air, Motor carrier (truck), Pipeline, Rail, Water
- Transportation mode performance is measured in (ton x miles) over a year
- Wide disparities in the various infrastructures exist between highly populated countries
- Lack of infrastructure makes it difficult to use that transportation mode domestically

## Table 12.1: Infrastructure Statistics in Several Countries

|                                       | Brazil     | China        | Germany    | Nigeria               | United States |
|---------------------------------------|------------|--------------|------------|-----------------------|---------------|
| Air <sup>a</sup>                      | 7          | 71           | 14         | 10                    | 189           |
| Highway<br>(paved)                    | 212,798 km | 3,453,890 km | 645,000 km | 28,980 km             | 4,304,715 km  |
| Pipeline (oil)                        | 4,831 km   | 23,072 km    | 2,826 km   | 4,441 km              | 240,711 km    |
| Broad gauge<br>(1.676 meters) rail    | 5,822 km   | 100 km       |            |                       |               |
| Standard gauge<br>(1.435 meters) rail | 194 km     | 190,000 km   | 43,209 km  | 293 km                | 293,564 km    |
| Narrow gauge<br>(1.000 meter) rail    | 23,341 km  | 670 km       | 220 km     | 3,505 km <sup>b</sup> |               |
| Water (inland)                        | 50,000 km  | 110,000 km   | 7,467 km   | 8,600 km              | 41,009 km     |

<sup>a</sup>Number of paved runways over 3,047 meters (approximately 10,000 feet).

<sup>b</sup>gauge = 1.067 meters

Source: The World Factbook, [www.cia.gov](http://www.cia.gov), 2016.

Rail gauge: distance between the inner sides of two parallel rail tracks

# Transportation Modes (1 of 16)

- The attractiveness of a particular mode depends on the following attributes:<sup>1</sup>
  - Cost (price that a carrier charges to transport a shipment)
  - Speed (elapsed transit time from pickup to delivery)
  - Reliability (consistency of delivery)
  - Capability (number of different types of product)
  - Capacity (volume that can be carried at one time)
  - Flexibility (ability to deliver the product to the customer)

<sup>1</sup>Drawn from David J. Bloomberg, Stephen LeMay, and Joe B. Hanna, *Logistics* (Upper Saddle River, NJ: Prentice Hall, 2002), Chapter 7.



# Transportation Modes (2 of 16)

- Airfreight
  - Generally the fastest mode for shipment exceeding 965 km per hour
  - Expensive
  - **Accessorial service**, if needed, adds transportation cost and time (pickup from supplier to airport and delivery from airport to destination)
  - Best suited for high-value, lower-volume urgent, perishable, or time-specific deliveries
  - Dimensional weight used for rates



All cargo airplane carries  
up 150 tons of freight

# Transportation Modes (3 of 16)

- Great majority of airfreight is carried in the freight compartments of passenger airplanes (limited capacity)
- Examples of products that move by airfreight:
  - Electronic or electrical equipment
  - Cut flowers and nursery stock of plants
  - Fruits and vegetables
  - Machinery and parts
  - Auto parts and accessories
  - Metal products
  - Photographic equipment, parts, and film
  - Printed matter
  - Wearing apparel



# Transportation Modes (4 of 16)

- Airfreight
  - Reliability is problematic due to delays caused by:
    - Congestion and resultant delays with air passenger transportation (belly freight in passenger airplanes)
    - Weather (e.g., fog, snow, thunderstorms)

# Transportation Modes (5 of 16)

- Motor carriers
  - Most important business user of the Interstate Highway System
  - Primary advantage is flexibility
  - Cost is generally lower when compared to airfreight
  - Types:
    - Less-than-truckload (LTL) carries shipment for multiple customers served in a route, example of carriers, Aramex, Samsa, Fedex freight, UPS freight
    - Full Truckload (TL) from customer to customer (consolidated shipment)

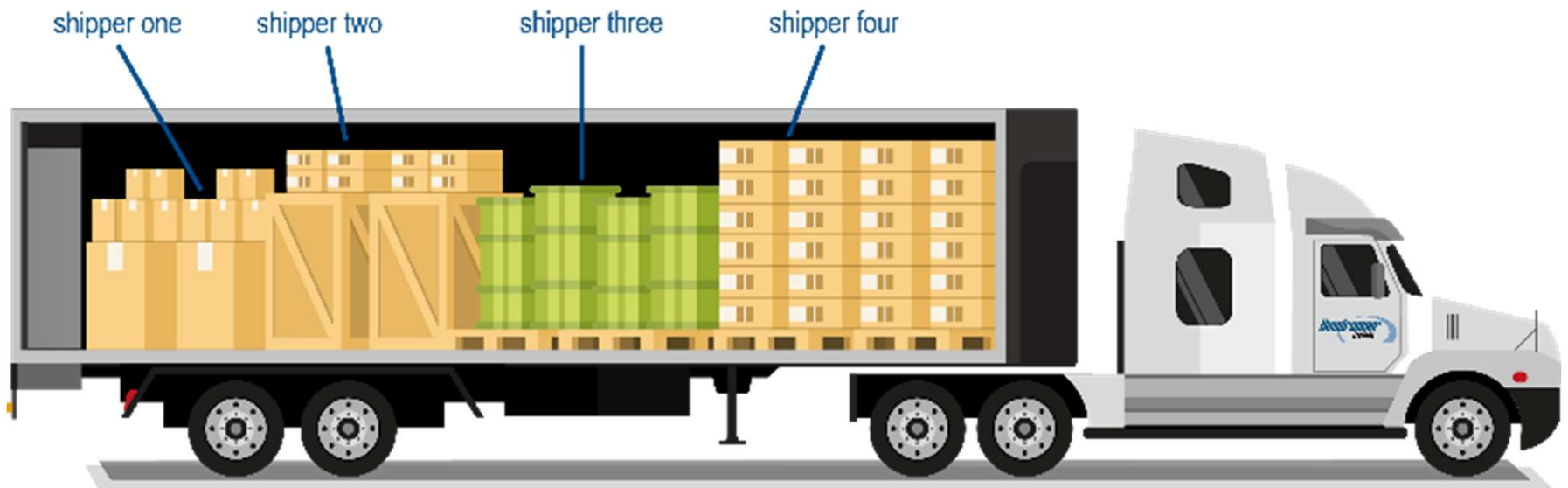


# Transportation Modes (6 of 16)

## Motor carriers

- Less-than-truckload (LTL)
  - 150 to 10,000 pounds
  - Too small to fill a truck
  - LTL trucks carry shipments from many shippers
  - Operate through a system of terminals and serve customers in route (milk run)
- Process
  1. Shipments are taken to a terminal, where they are loaded to line-haul trucks
  2. Line-haul to terminal near destination
  3. Shipments loaded to smaller trucks for local delivery
  4. From each terminal small trucks go out to customers, delivering and picking up shipments

## Less-than-Truckload (LTL) Shipment



The trailer is filled up with pallets from individual shippers who are paying for space based on the dimensional space of their goods.

# Transportation Modes (8 of 16)

- Motor carriers
  - Truckload (TL)
    - Focus on shipments more than 10,000 pounds
    - Exact weight depends on the product
    - Close to the amount that would physically fill a truck trailer
    - Possible that large shipments from several customers can be consolidated

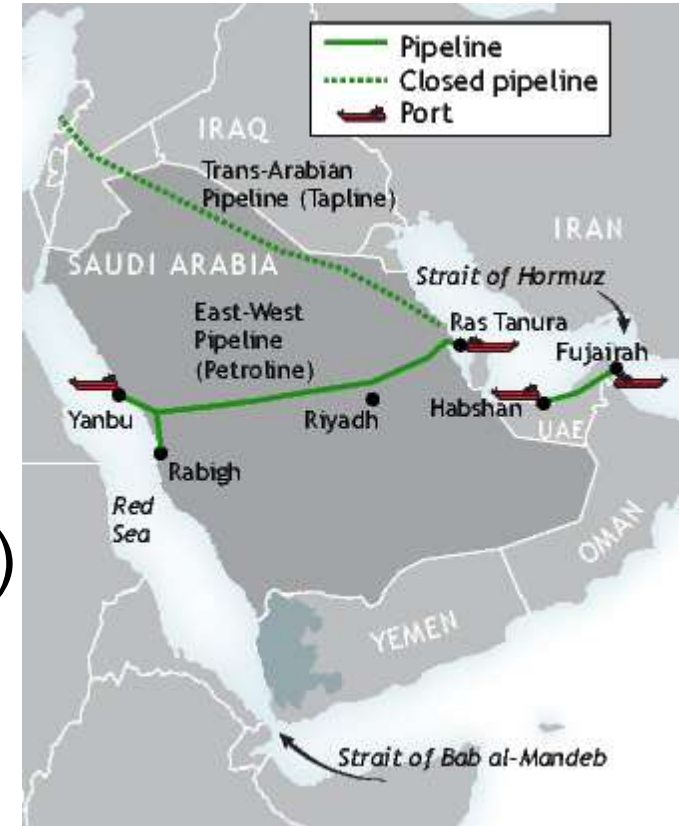
# Transportation Modes (9 of 16)

- Truckload (TL) Motor carriers
  - Prominent TL carriers include:
    - J.B. Hunt, Schneider National, Swift Transportation, Werner Enterprises
    - Global trucking terminal (Dammam), Al Muhaidib Land Transport, Nesma, Globe Trucking Jubail
  - Process
    - Shipments tend to move directly from the shipper's location to the consignee's location
  - Involve speed limits and hours-of-service (HOS) rules. HOS rules are subject of constant legislation (job standard)



# Transportation Modes (11 of 16)

- Pipelines
  - Only mode without vehicles
  - No need for vehicle operators
  - Transportation is one way
  - Most reliable mode (no traffic accident, delay to weather conditions)
- Characteristics
  - Tend to be the slowest mode
  - Accommodates liquid, liquefiable, or gaseous products
  - Capable of transporting large product volumes
  - High fixed costs, but relatively low cost per unit due to large product volume

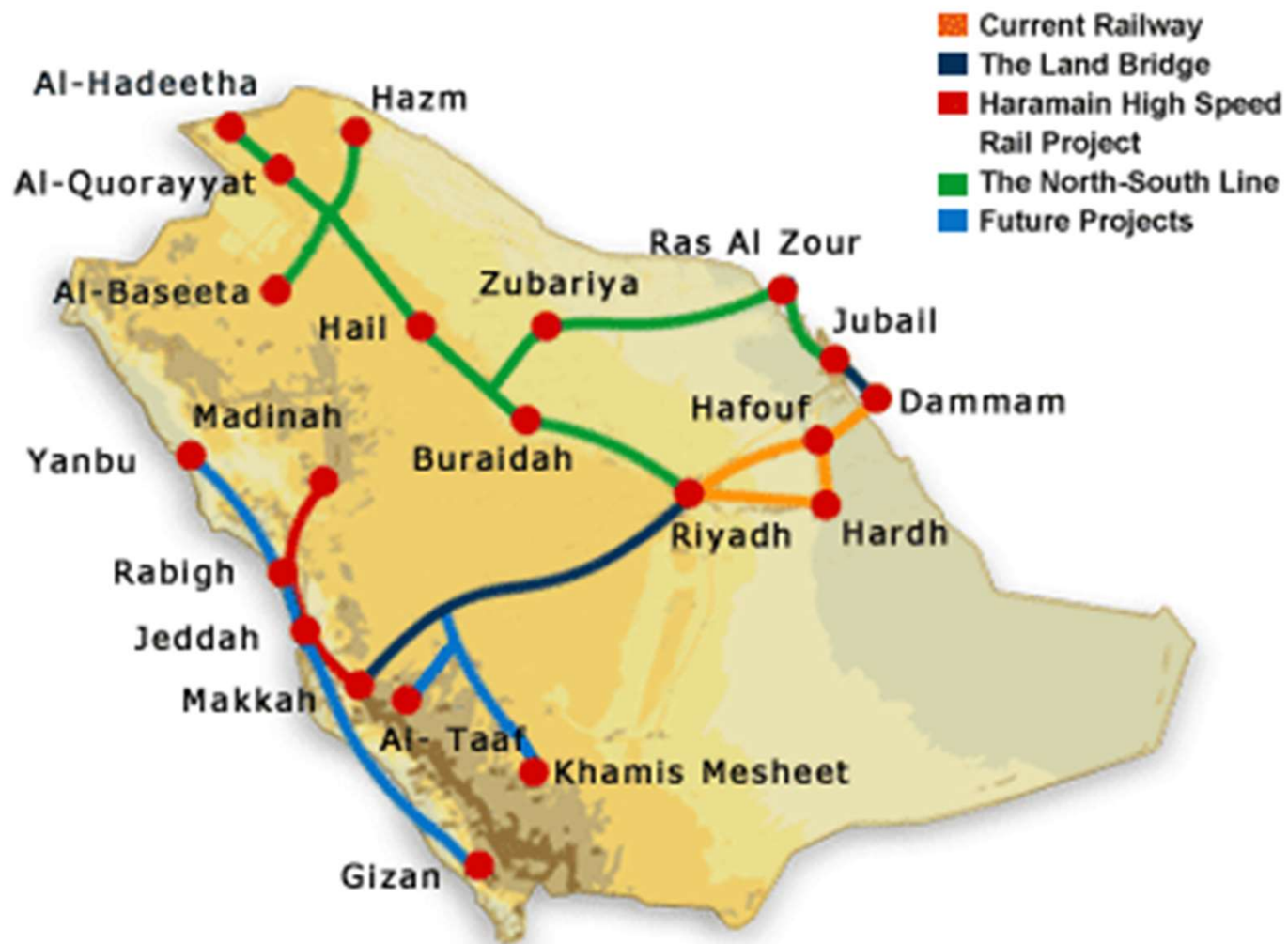


# Transportation Modes (13 of 16)

- Railroads
  - U.S. dominated by four freight carriers:
    - Burlington Northern (west of the Mississippi)
    - CSX (east of the Mississippi)
    - Norfolk Southern (east of the Mississippi)
    - Union Pacific (west of the Mississippi)
  - Domination can limit service and pricing options



Jeddah-Riyadh Rail



# Transportation Modes (14 of 16)

- Railroads
  - Neither “best” or “worst” on any of the six attributes
  - Superior to air, motor, and pipeline, but inferior to water in ability to transport different kinds of products
  - Less flexibility than motor carriers, but more when compared to air, water, and pipeline
  - Flexibility due by different types of rail cars:
    - Boxcars (general container freight)
    - Hopper cars (coals and minerals)
    - Tank cars (liquid or liquefiable product)
    - Scaling capacity by adding more cars as need up to a limit





# Transportation Modes (15 of 16)

- Railroads
  - Superior to air and motor with regards to volume, but inferior to pipeline and water
  - Less expensive than air and motor, but more expensive than pipeline and water
  - Faster than pipeline and water, but slower than air and truck
  - Growth of intermodal transportation has given railroads access to manufactured and packaged products, which tend to be higher value

# Transportation Modes (16 of 16)

- Water (inland waterways vs oceanic waterways)
  - Somewhat unreliable
    - Drought
    - Icing
    - Aging lock system
  - Slow average speeds
  - Relatively inexpensive
  - Focuses on lower value bulk commodities handled by mechanical means
  - Many different kinds of products can be carried
  - Carries greater volumes than rail or truck up to 28,000 containers



# Comparison of mode of transport

| Characteristics      | Rail      | Road   | Pipeline  | Air       | Water     |
|----------------------|-----------|--------|-----------|-----------|-----------|
| Door to door service | Sometimes | Yes    | Sometimes | No        | Sometimes |
| Price                | Low       | High   | Very low  | Very high | Very low  |
| Speed                | Slow      | Medium | Slow      | Very fast | Very slow |
| Reliability          | Medium    | Medium | Very high | Very high | Low       |
| Packaging needs      | High      | Medium | Nil       | High      | High      |
| Risk of loss/damage  | Medium    | Medium | Very low  | Low       | Medium    |
| Flexibility          | Low       | High   | Very low  | Very low  | Low       |
| Environmental impact | Low       | Medium | Low*      | High      | Low       |

\* Possibility of disaster causing significant impact

Adapted from Gourdin K, Global Logistics Management 2006



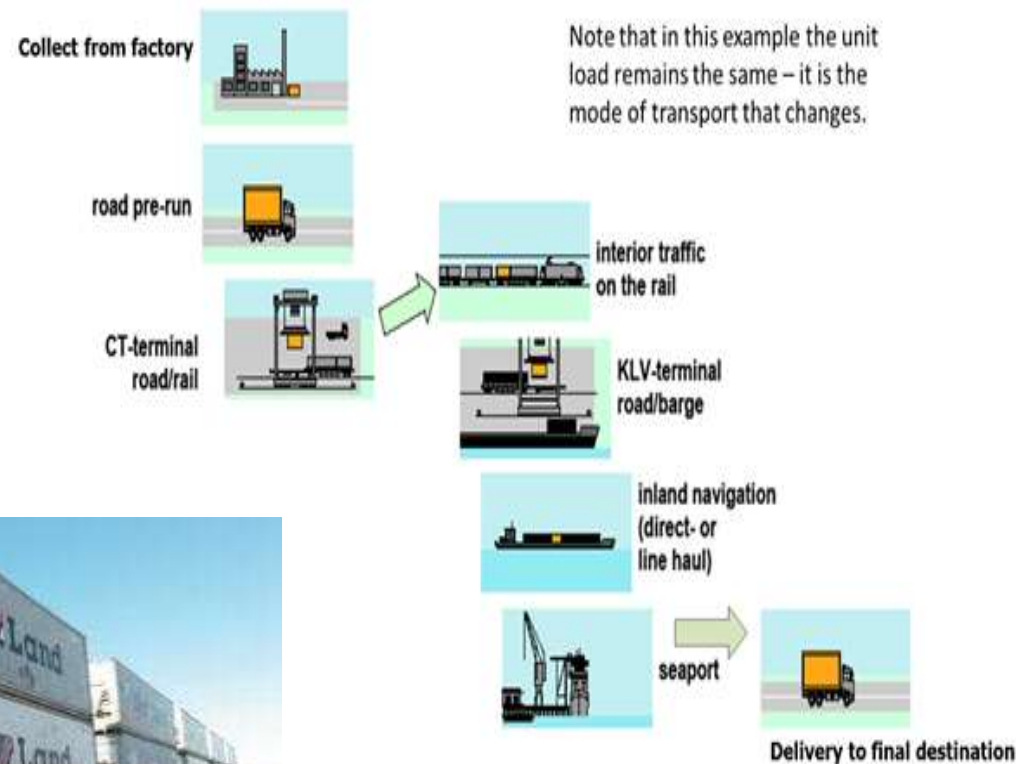
# Intermodal Transportation (1 of 4)

- Intermodal transportation
  - Refers to transportation when using a container or other equipment that can be transferred from the vehicle of one mode to the vehicle of another mode without the contents being reloaded or disturbed<sup>2</sup>

<sup>2</sup>This definition comes from Barton Jennings and Mary C. Holcomb, “Beyond Containerization: The Broader Concept of Intermodalism,” *Transportation Journal* 35, no. 3 (1995): 5–13.

# Intermodal Transportation (2 of 4)

- Intermodal transportation
  - Two or more modes work closely together to utilize advantages of each mode while minimizing their disadvantages
  - Example:
    - Piggyback transportation
      - Trailer-on-flatcar
      - Container-on-flatcar



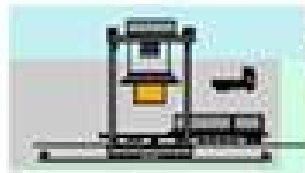
**Collect from factory**



**road pre-run**



**CT-terminal  
road/rail**



Note that in this example the unit load remains the same – it is the mode of transport that changes.



**interior traffic  
on the rail**



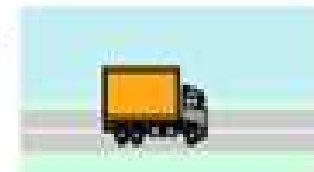
**KLV-terminal  
road/barge**



**inland navigation  
(direct- or  
line haul)**



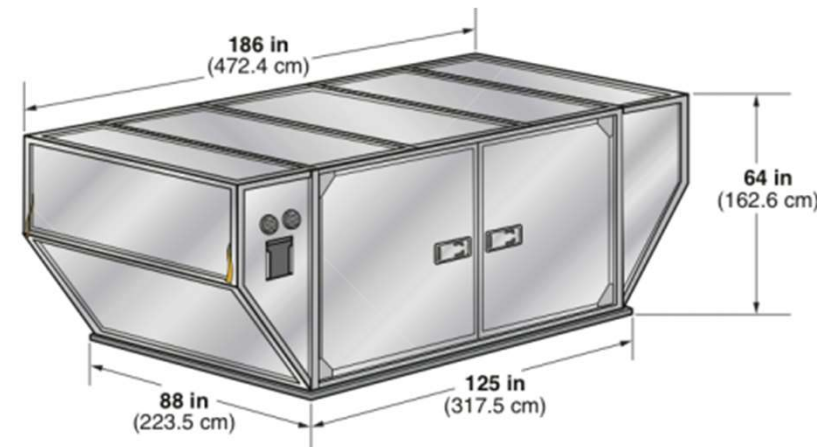
**seaport**



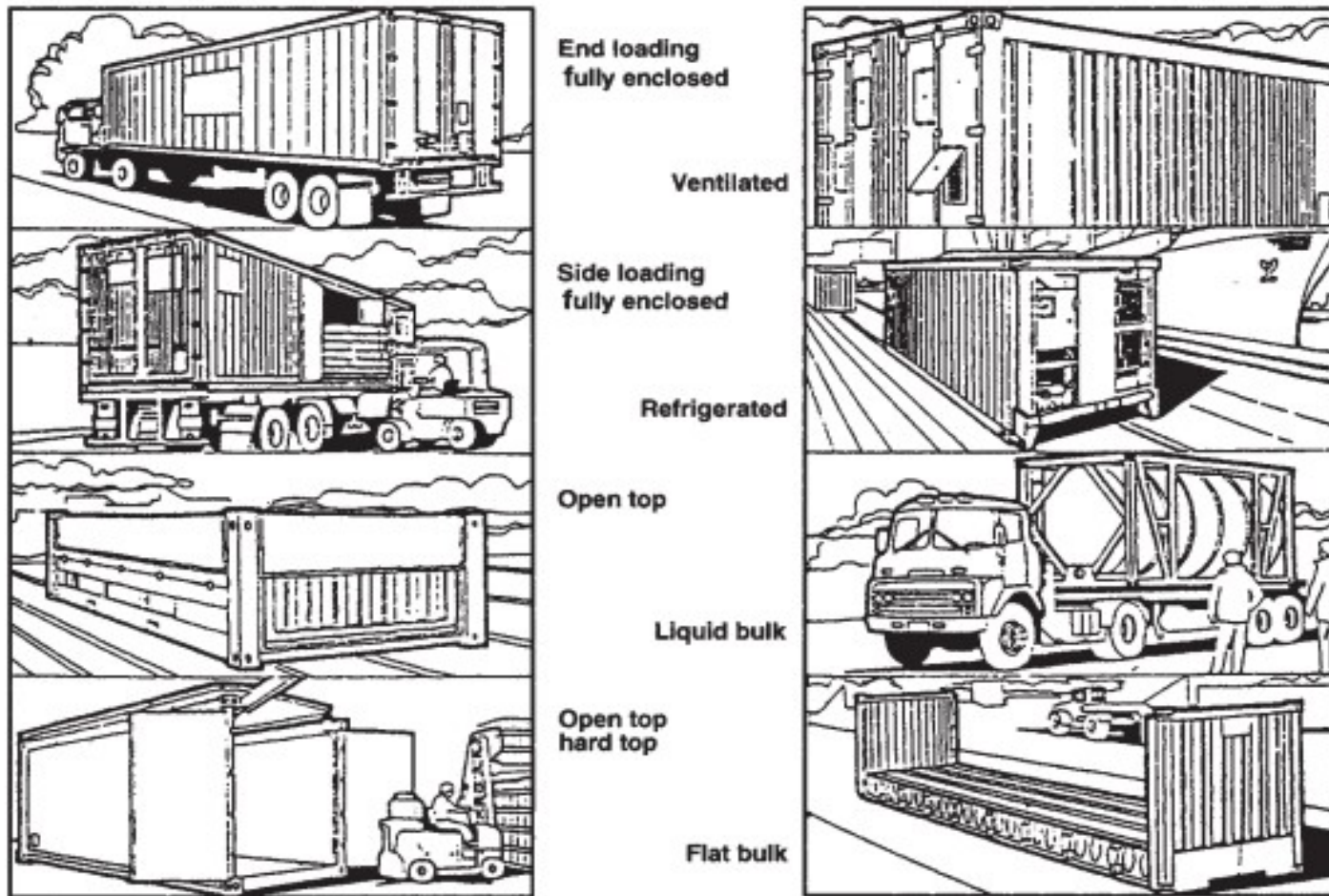
**Delivery to final destination**

# Intermodal Transportation (3 of 4)

- Containers
  - Moved by mechanical devices (e.g., container crane)
  - Provide significant reduction in freight handling costs
  - Are interchangeable among rail, truck, and water carriers
  - Are measured by TEUs (20-foot equivalent unit)
  - Allowed for land bridge services (sea-land-sea) that reduces transport time
  - Airfreight containers (unit load devices, or ULDs) are designed specifically for fuselage



# Figure 12.1: Various Types of Intermodal Surface Containers



**Unit load:**  
**twenty-foot**  
**equivalent unit**  
**(TEU)**  
20-foot unit load.  
For example:  
sea vessels has  
capacity of  
15,000 TEUs

<http://www.thefalconfreight.com/container-sizes/>

Source: Ports of the World, 14th edition, CIGNA Property & Casualty Companies.

# Transportation Specialists (1 of 6)

- Freight forwarders (not a new mode but offering value added services)
  - Two types of domestic freight forwarders:
    - Surface
    - Air
  - Exists by offering a service to shippers that must use LTL rates (expensive rate) because they do not generate enough volume to use TL rates (cheaper rate)
  - Taking advantage of load consolidation from different customers
  - Typically offers pickup and delivery service but does not perform the line-haul service (done by motor carriers or railroads)

# Transportation Specialists (2 of 6)

- Freight forwarders
  - Give volume discounts to customers shipping large quantities of freight at one time
  - TL rates less than LTL rates because
    - Shipper loads the goods, and consignee unloads trailer
    - The load goes directly from shipper to consignee without passing through terminals
    - Paperwork, billing, and other administrative costs are little more for a 25,000-pound shipment than for 250-pound shipment



# Transportation Specialists (3 of 6)

- Air forwarders
  - Consolidate shipments
  - Tender to airlines in containers ready for loading
  - Forwarders provide retailing function (like service for small shipment)
  - Airline provides wholesaling function (like service for ULD shipment)
- Shippers' associations
  - Similar to air and freight forwarders but are not-for-profit organizations
  - Primarily focused on achieving the lowest rates for members

# Transportation Specialists (4 of 6)

- Brokers
  - Can handle both LTL and TL
  - Companies that look to match a shipper's freight with a carrier to transport it
  - Those handling LTL consolidate and give them to motor carriers, freight forwarders, or shippers' associations
  - If handling TL shipments, brokers retain a carrier and receive a portion of transportation charges
- Third-party logistics companies (3PLs)
  - Find clients with complimentary transportation needs to maximize equipment utilization
  - Should reduce transportation costs to clients

# Transportation Specialists (6 of 6)

- Parcel carriers
  - Parcels are packages weighing up to 150 pounds (68 KG)
  - Parcel carriers are companies that specialize in transporting parcels
  - Parcel carriers include:
    - U.S. Postal Service – Retail Ground
    - United Parcel Service (UPS)
    - FedEx Express
    - Greyhound Package Express

# Transportation Regulation (1 of 8)

- The five modes are influenced by governmental regulations
  - Examples:
    - Mandatory retirement age for pilots in U.S.
    - Placement of lighting on truck trailers
- Regulation
  - Costs money
  - Needs to be codified (standardized)
  - Is enforced by government agencies
- Regulation are related to: environment (noise, air pollution, resource preservation), safety (transportation), economic (regulation vs deregulation)

# Transportation Regulation (2 of 8)

- Level and degree of regulation varies from country to country
  - For example, industrialized countries tend to have more stringent transportation equipment emissions regulations when compared to those of less industrialized countries
- Logisticians must understand:
  - Relevant transportation regulations
  - Cost and service implications of regulations

# Transportation Regulation (3 of 8)

- Environmental regulation
  - Environment Protection Agency (EPA) is the federal regulatory agency established to protect human health and the environment
  - Current concerns include:
    - Noise and air pollution
    - Resource conservation

# Transportation Regulation (4 of 8)

- Safety regulation
  - Department of Transportation (DOT) is the federal agency responsible for transportation safety regulations for all five modes
  - Federal Aviation Administration (FAA) has primary responsibility for air transportation safety
  - Federal Motor Carrier Safety Administration (FMCSA) is focused on reducing crashes, injuries, and fatalities involving large trucks and buses



# Transportation Regulation (5 of 8)

- Safety regulation
  - Office of Pipeline Safety (OPS) is responsible for safety considerations for natural gas and liquid pipelines
  - Federal Railroad Administration (FRA) has primary responsibility for safety in the U.S. railroad industry
  - U.S. Coast Guard (USCG) has three broad roles associated with marine (water) transportation:
    - Safety
    - Security
    - Stewardship

# Transportation Regulation (6 of 8)

- Economic regulation
  - Refers to control over business practices and activities such as entry and exit, pricing, service, accounting and financial issues, and mergers and acquisitions
  - Regulation began in the 1870s due to a belief that transportation companies would not act in the public's best interest without government regulation<sup>3</sup>

<sup>3</sup>Donald V. Harper, *Transportation in America*, 2nd ed. (Englewood Cliffs, NJ: Prentice Hall, 1982), Chapter 19.

# Transportation Regulation (7 of 8)

- Economic regulation
  - Resulted in two key economic regulatory bodies:
    - Interstate Commerce Commission (ICC)
    - Civil Aeronautics Board (CAB)

# Transportation Regulation (8 of 8)

- Economic regulation
  - Deregulation began in the 1970s
    - CAB was eliminated in 1985
    - ICC was legislated out of existence in 1995
  - A new agency, the Surface Transportation Board (STB), was created
  - Economic deregulation has allowed greater freedom with respect to pricing and service options

# Legal Classification of Carriers (1 of 3)

- Transportation carriers are classified as either:
  - For hire
    - Common
    - Contract
    - Exempt
  - Private
- Classification is important because different levels of economic regulation are applicable to different carriers

# Legal Classification of Carriers (2 of 3)

- For-hire carriers
  - Common carriers
    - Serve the general public
  - Contract carriers
    - Offer specialized service to customers on a contractual basis
    - No obligation to serve the general public or to treat customers on an equal basis

# Legal Classification of Carriers (3 of 3)

- For-hire carriers
  - Exempt carriers
    - Exempted from economic regulation due to legislation
- Private carriers
  - Companies whose primary business is other than transportation and provide their own transportation service
  - Also exempt from economic regulation

# Transportation Optimization

## Freight traffic assignment problem (TAP)

1. Freight traffic assignment problem (TAP)
  1. Determine a least-cost routing of goods over an existing network from origins (e.g plants) to destinations (e.g retailers)
  2. Modelled as network flow problem
  3. Can be static if decisions are not affected explicitly by time
  4. Can be dynamic if time dimension is explicitly taken into account
  5. Single vs multi-commodity type



# Transportation Optimization

## Freight traffic assignment problem (TAP)

Case single commodity, static, linear TAP:

- NTN is a Swiss freight intermodal carrier located in Lausanne that supplies empty containers (EC) to customers and transports them to destinations
- NTN needs to optimize weekly empty containers (EC) reallocation. Empty container transport cost is expensive
- Several empty ECs had to be reallocated among the existing terminals
- $o_j$  amount of EC's to move from terminal  $j$
- $d_j$  demand of EC's at terminal  $j$
- $c_{ij}$  cost of transporting EC from terminal  $i$  to  $j$

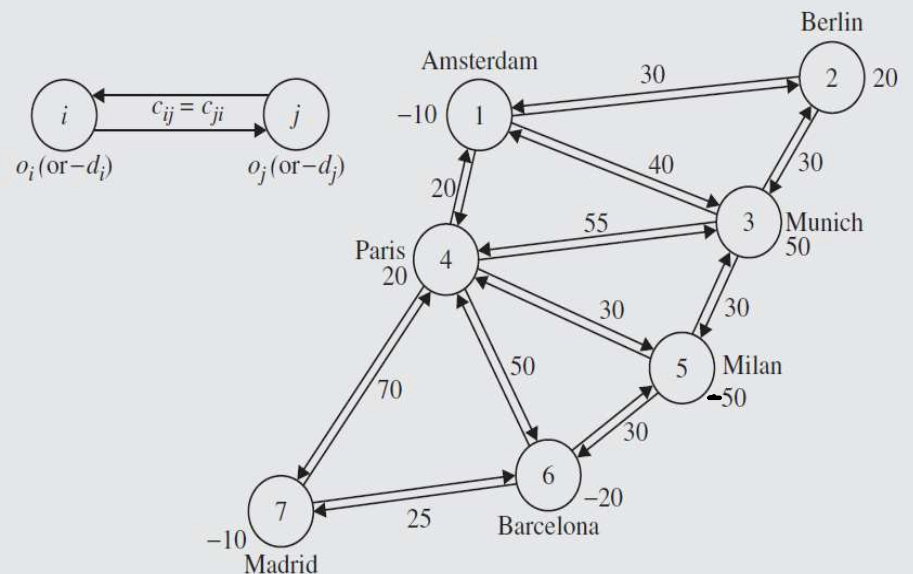


Figure 6.10 Graph representation of the NTN empty container allocation problem.

# Freight traffic assignment problem (TAP)

## Formulation NTN Case

- **Input:** Terminal  $i \in \{1, 2, \dots, 7\}$ ,  $o_j$  amount of EC's to move from terminal  $j$ ,  $d_j$  demand of EC's at terminal  $j$ ,  $c_{ij}$  cost of transporting EC from terminal  $i$  to  $j$
- **Decision variables:** how many ECs to move from terminal  $i$  to  $j$ ,  $x_{ij}$  (non-negative numbers)
- **Objective:** Minimize total transportation cost of EC's

$$\sum_{i=1}^7 \sum_{j=1}^7 c_{ij} x_{ij}$$

- **Constraint 1:** Satisfy demand of terminal  $j$ , Input flow to  $j$  equal to output flow of  $j$  plus  $d_j$   $\sum_{i=1}^7 x_{ij} = \sum_{i=1}^7 x_{ji} + d_j$
- **Constraint 2:** Satisfy # of EC's to move from  $j$ , input flow equal out flow of  $j$  minus  $o_j$ :  $\sum_{i=1}^7 x_{ji} = \sum_{i=1}^7 x_{ij} - o_j$

# Transportation Optimization

## NTN case optimal results (TAP)

What if supply is smaller than demand?

|           | Transportation cost per EC (Euros) |           |           |           |           |           |           |                           |
|-----------|------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|---------------------------|
|           | Amsterdam                          | Berlin    | Munich    | Paris     | Milano    | Barcelona | Madrid    | Demand (-)<br>/Supply (+) |
| Amsterdam | 1000                               | <b>30</b> | <b>40</b> | <b>20</b> | 1000      | 1000      | 1000      | -10                       |
| Berlin    | <b>30</b>                          | 1000      | <b>30</b> | 1000      | 1000      | 1000      | 1000      | 20                        |
| Munich    | <b>40</b>                          | <b>30</b> | 1000      | <b>55</b> | <b>30</b> | 1000      | 1000      | 50                        |
| Paris     | <b>20</b>                          | 1000      | <b>55</b> | 1000      | <b>30</b> | <b>50</b> | <b>70</b> | 20                        |
| Milano    | 1000                               | 1000      | <b>30</b> | <b>30</b> | 1000      | <b>30</b> | 1000      | -50                       |
| Barcelona | 1000                               | 1000      | 1000      | <b>50</b> | <b>30</b> | 1000      | <b>25</b> | -20                       |
| Madrid    | 1000                               | 1000      | 1000      | <b>70</b> | 1000      | <b>25</b> | 1000      | -10                       |

|                | Optimal results Decision variables x <sub>ij</sub> # of EC's moved from terminal i to j |        |        |       |        |           |        |             |                            |
|----------------|-----------------------------------------------------------------------------------------|--------|--------|-------|--------|-----------|--------|-------------|----------------------------|
|                | Amsterdam                                                                               | Berlin | Munich | Paris | Milano | Barcelona | Madrid | Output flow | Output - demand (- supply) |
| Amsterdam      | 0                                                                                       | 0      | 0      | 0     | 0      | 0         | 0      | 0           | 10                         |
| Berlin         | 10                                                                                      | 0      | 10     | 0     | 0      | 0         | 0      | 20          | 0                          |
| Munich         | 0                                                                                       | 0      | 0      | 0     | 60     | 0         | 0      | 60          | 10                         |
| Paris          | 0                                                                                       | 0      | 0      | 0     | 0      | 10        | 10     | 20          | 0                          |
| Milano         | 0                                                                                       | 0      | 0      | 0     | 0      | 10        | 0      | 10          | 60                         |
| Barcelona      | 0                                                                                       | 0      | 0      | 0     | 0      | 0         | 0      | 0           | 20                         |
| Madrid         | 0                                                                                       | 0      | 0      | 0     | 0      | 0         | 0      | 0           | 10                         |
| 1-5 Input flow | 10                                                                                      | 0      | 10     | 0     | 60     | 20        | 10     | Opt cost    | <b>3,900</b>               |

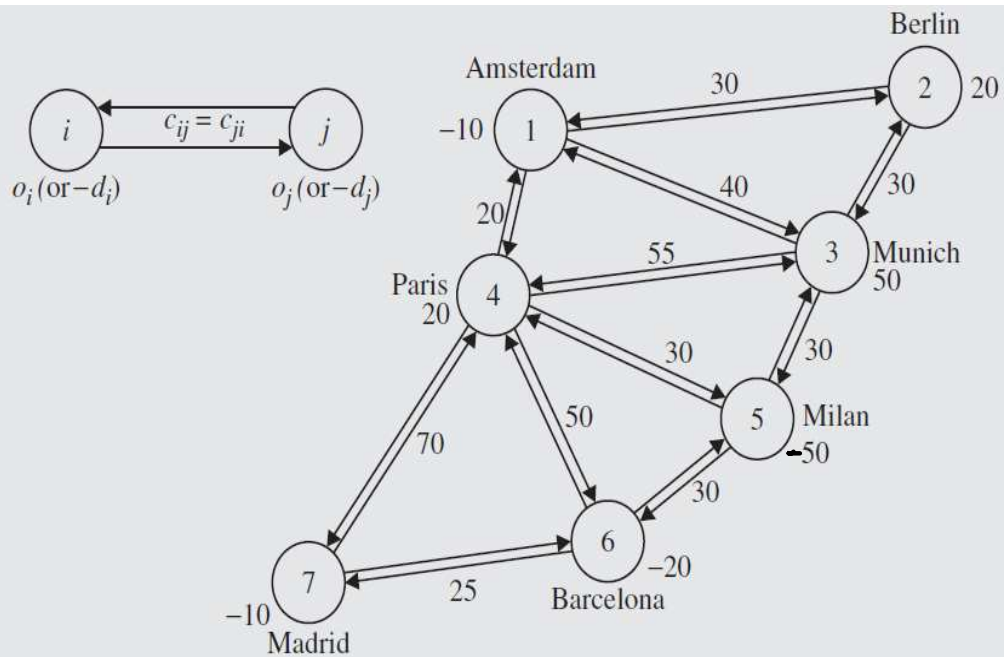
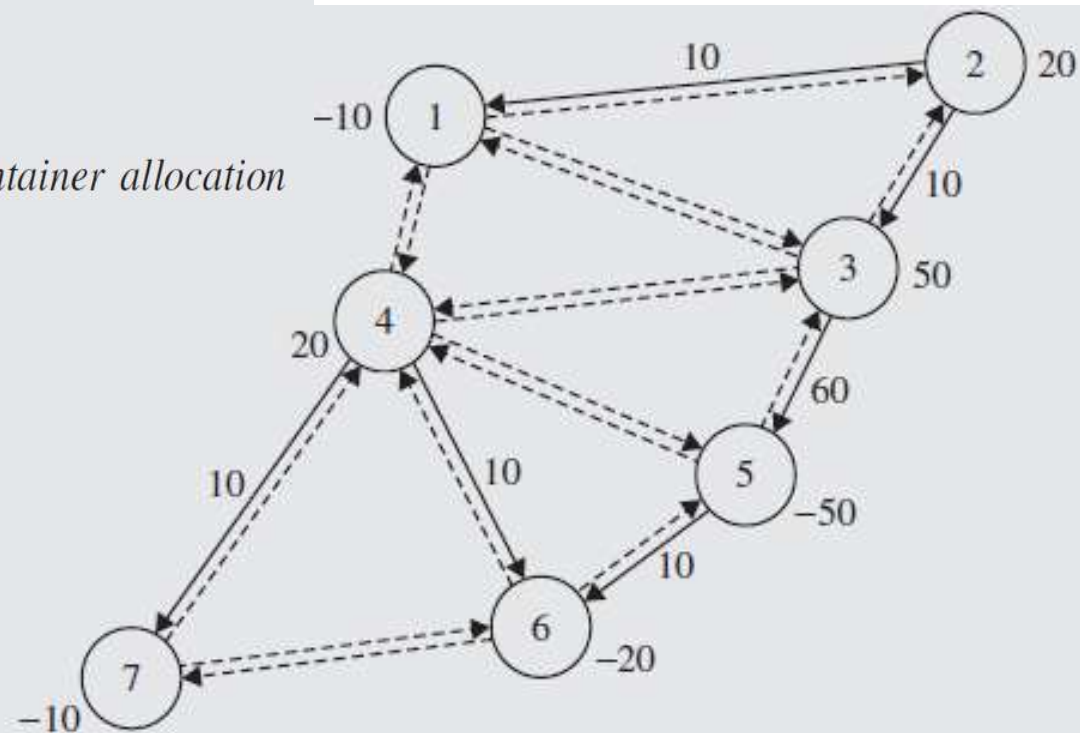


Figure 6.10 Graph representation of the NTN empty container allocation problem.



# Service network design problem (SNDP)

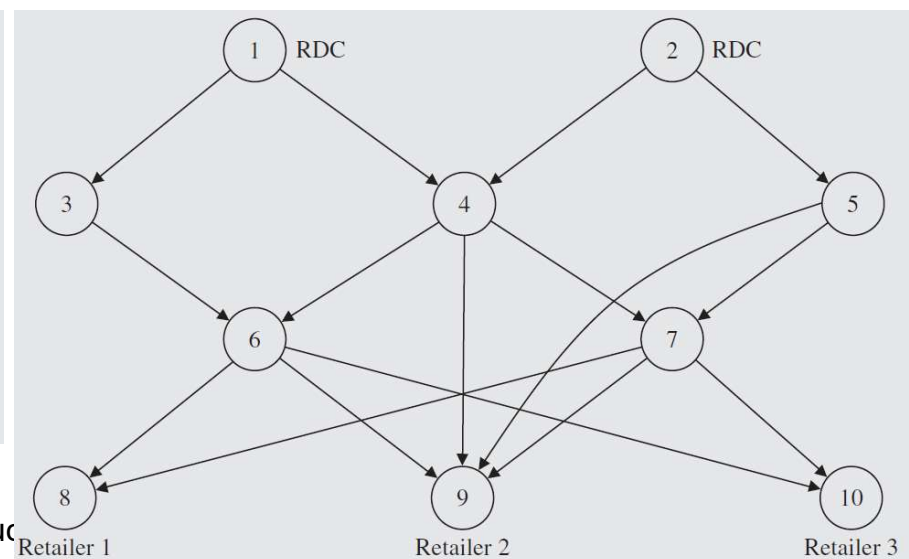
- Relevant in case of consolidation based transportation carriers with cross docking terminals
- Objective minimize cost taking into carrier's operating costs and demand expectations
- Basic SNDP is when a fixed cost is to be paid for link  $i$  to  $j$ , called fixed charge network design. Aim to find:
  - Which links have to be employed?
  - How much goods to transport on the selected links?

# Service network design problem (SNDP):

## Case FHL fast carrier delivery

- FHL is fast carrier with core business transport of palletized load units
- Transportation service is carried through five cross-docking facilities
- A customer needs to move pallets from two RDCs to three retailers
- Transportation is realized by FHL between each pair of facilities using a single truck with load up to 200 pallets. Only when the terminal is the retailer, FHL uses larger trucks of 250-pallet capacity.
- Pallets are classified into three categories, liquid, dry food, and others
- Average retailer daily demand in the next trimester is given
- Pallets are available each day at the RDC in the quantities indicated

| Category    | Retailer |    |    | RDC |     |
|-------------|----------|----|----|-----|-----|
|             | 1        | 2  | 3  | 1   | 2   |
| Liquid food | 80       | 55 | 40 | 80  | 95  |
| Dry food    | 74       | 52 | 64 | 90  | 100 |
| Other       | 47       | 42 | 76 | 75  | 90  |



# Service network design problem (SNDP):

## Case FHL fast carrier delivery

- **Input parameters:**  $f_{ij}$  fixed cost to establish link  $i, j$ ,  $c_{ij}^k$  cost of transporting a pallet from  $i, j$  of type  $k$ ,  $u_{ij}$  capacity of link  $i, j$
- **Decision variables:** links to be employed,  $y_{ij}$  (**binary**), number of pallets transported from terminal  $i$  to  $j$  of type  $k$ ,  $x_{ij}^k$  (**nonnegative**)
- **Objective:** minimize total cost variable and fixed cost
$$\sum_i \sum_j \sum_k c_{ij}^k x_{ij}^k + \sum_i \sum_j f_{ij} y_{ij}$$
- **Constraint 1:** Satisfy demand of retailer  $j$ , Input flow to  $j$  equal  $d_j$ ,
$$\sum_i x_{ij}^k = d_j^k, \forall j, k$$
- **Constraint 2:** Satisfy RDC  $j$  capacity, output flow of RDC  $j$  equal to  $o_j$ :
$$\sum_i x_{ij}^k = o_j^k, \forall j, k$$
- **Constraint 3:** Input flow of crossdocking terminal  $j$  is equal to output flow  $\sum_i x_{ij}^k = \sum_i x_{ji}^k, \forall j, k$
- **Constraint 4:** Unmet link capacity is nonnegative  $u_{ij} y_{ij} - \sum_k x_{ij}^k \geq 0$



# Service network design problem (SNDP): Case FHL fast carrier delivery Optimal

|                  | Variable cost |            |            |          |
|------------------|---------------|------------|------------|----------|
| link (i,j)       | $c_{ij}^1$    | $c_{ij}^2$ | $c_{ij}^3$ | $f_{ij}$ |
| <b>(RDC 1,3)</b> | 22            | 19         | 17         | 450      |
| <b>(RDC 1,4)</b> | 22            | 19         | 17         | 490      |
| <b>(RDC 2,4)</b> | 20            | 21         | 18         | 390      |
| <b>(RDC 2,5)</b> | 20            | 21         | 18         | 435      |
| ⋮                | ⋮             | ⋮          | ⋮          | ⋮        |
| <b>(6,R2)</b>    | 24            | 25         | 25         | 490      |
| <b>(6,R3)</b>    | 24            | 25         | 25         | 725      |
| <b>(7,R1)</b>    | 26            | 24         | 27         | 710      |
| <b>(7,R2)</b>    | 24            | 24         | 25         | 525      |
| <b>(7,R3)</b>    | 23            | 23         | 24         | 410      |

|                  | Decision Variable Optimal |            |            |          |
|------------------|---------------------------|------------|------------|----------|
| link (i,j)       | $x_{ij}^1$                | $x_{ij}^2$ | $x_{ij}^3$ | $y_{ij}$ |
| <b>(RDC 1,3)</b> | 79                        | 74         | 47         | 1        |
| <b>(RDC 1,4)</b> | 1                         | 16         | 28         | 1        |
| <b>(RDC 2,4)</b> | 55                        | 36         | 14         | 1        |
| <b>(RDC 2,5)</b> | 40                        | 64         | 76         | 1        |
| ⋮                | ⋮                         | ⋮          | ⋮          | ⋮        |
| <b>(6,R2)</b>    | 0                         | 0          | 0          | 0        |
| <b>(6,R3)</b>    | 0                         | 0          | 0          | 0        |
| <b>(7,R1)</b>    | 0                         | 0          | 0          | 0        |
| <b>(7,R2)</b>    | 0                         | 0          | 0          | 0        |
| <b>(7,R3)</b>    | 40                        | 64         | 76         | 1        |

Optimal total cost: 34532

See FHL fast carrier Excel file for further details



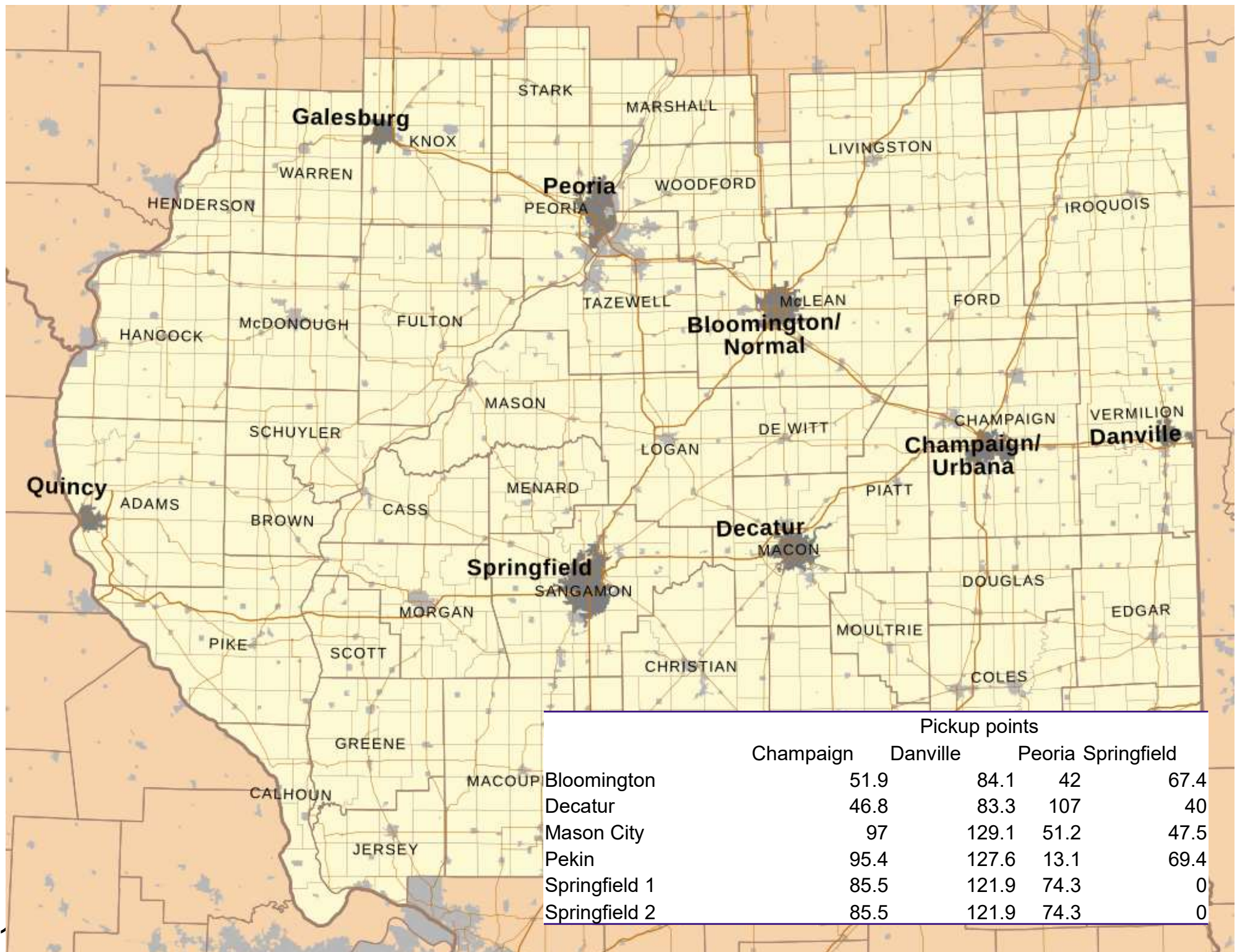
# Dynamic driver assignment problem

- The focus is on full Truck load trips to be assigned to drivers
- A trip takes several days
- Driver is assigned to single trip at a time
- $D$  is the set of drivers, and  $L$  be the set of transport services to be performed
- $c_{ij}$  is the cost to assigned driver  $i$  to trip  $j$ . it includes:
  - Additional cost to return the driver home after delivery completion

# Dynamic driver assignment problem

## Planet Transport Case

- Planet Transport is motor carrier in US that specializes in TL trucking.
- Last January 26, the company had to solve the problem of assigning four full-load transport services each of three days long.
- The pickup points were in Champaign, Danville, Peoria and Springfield.
- At that time six drivers were available. The first four were located in Bloomington, Decatur, Mason City and Pekin, and the last two in Springfield.
- Cost assignment of driver to a trip is proportional to distance between driver location and full-load trip pickup points



|               | Pickup points |          |        |             |
|---------------|---------------|----------|--------|-------------|
|               | Champaign     | Danville | Peoria | Springfield |
| Bloomington   | 51.9          | 84.1     | 42     | 67.4        |
| Decatur       | 46.8          | 83.3     | 107    | 40          |
| Mason City    | 97            | 129.1    | 51.2   | 47.5        |
| Pekin         | 95.4          | 127.6    | 13.1   | 69.4        |
| Springfield 1 | 85.5          | 121.9    | 74.3   | 0           |
| Springfield 2 | 85.5          | 121.9    | 74.3   | 0           |

# Dynamic driver assignment problem

## Planet Transport Case Formulation

- **Input parameters:**  $c_{ij}$  cost of assigning driver  $i$  to trip  $j$ ,  $D$  set of drivers (6 drivers),  $L$  set of trips (4 trips)
- **Decision variables:** driver  $i$  assignment to trip  $j$ ,  $x_{ij}$  (**binary**),  
**Objective:** minimize total variable

$$\sum_i \sum_j c_{ij} x_{ij},$$

- **Constraint 1:** Driver  $i$  is assigned to at most one trip

$$\sum_j x_{ij} \leq 1, \forall i \text{ in } D,$$

- **Constraint 2:** All trips should be assigned

$$\sum_i x_{ij} = 1, \forall j \text{ in } L,$$

Model Features:

- a. This problem is Binary linear problem.
- b. To solve it you can use OpenSolver as Excel Add in. (Solver ok)
- c. For this to be feasible number of trips to complete should be at least equal to number of drivers.



# Dynamic driver assignment problem

## Planet Transport Case Solution

| Input parameters |               | Pickup points |          |        |             |
|------------------|---------------|---------------|----------|--------|-------------|
|                  |               | Champaign     | Danville | Peoria | Springfield |
| Driver locations | Bloomington   | 51.9          | 84.1     | 42     | 67.4        |
|                  | Decatur       | 46.8          | 83.3     | 107    | 40          |
|                  | Mason City    | 97            | 129.1    | 51.2   | 47.5        |
|                  | Pekin         | 95.4          | 127.6    | 13.1   | 69.4        |
|                  | Springfield 1 | 85.5          | 121.9    | 74.3   | 0           |
|                  | Springfield 2 | 85.5          | 121.9    | 74.3   | 0           |

| Decision variables |               | Pickup points |          |        |             | Auxilliary       |
|--------------------|---------------|---------------|----------|--------|-------------|------------------|
|                    |               | Champaign     | Danville | Peoria | Springfield | Assigned drivers |
| Driver locations   | Bloomington   | 0             | 1        | 0      | 0           | 1                |
|                    | Decatur       | 1             | 0        | 0      | 0           | 1                |
|                    | Mason City    | 0             | 0        | 0      | 0           | 0                |
|                    | Pekin         | 0             | 0        | 1      | 0           | 1                |
|                    | Springfield 1 | 0             | 0        | 0      | 0           | 0                |
|                    | Springfield 2 | 0             | 0        | 0      | 1           | 1                |
| Auxilliary         | Unmet trips   | 0             | 0        | 0      | 0           |                  |

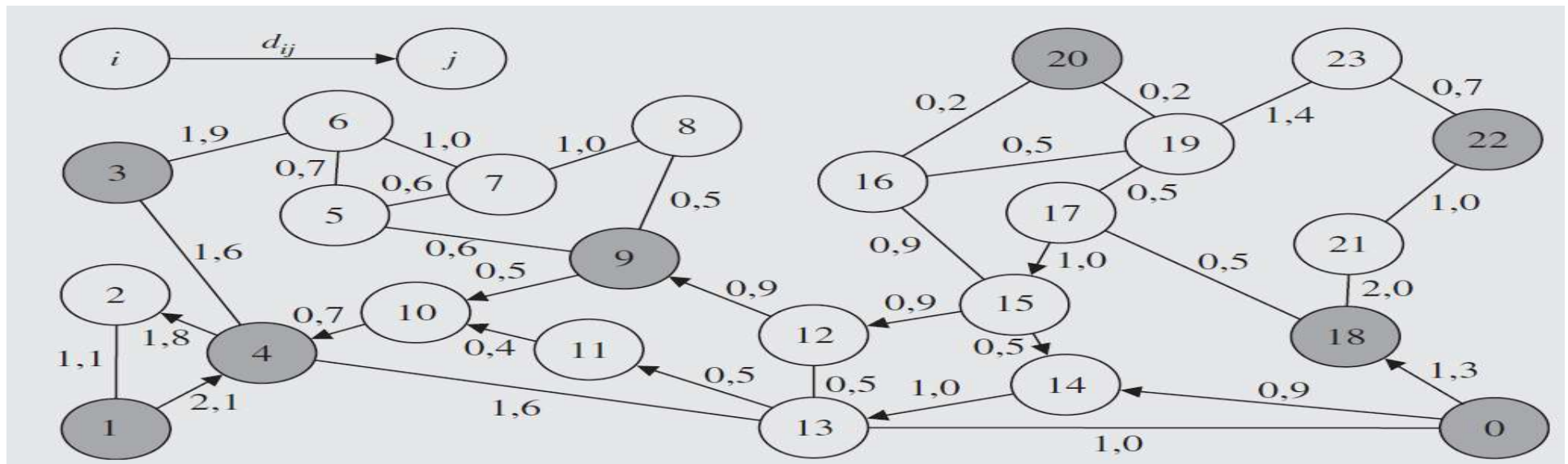
Optimal cost: 144

# Travel Salesman Problem (TSP)

- Aim at finding optimal delivery (collection) from a depot to numbers users
- One vehicle is used for the delivery
- Apply the milk-run concept, vehicle starts at depot serves all users and then returns back to depot (tour or circuit)
- In the tour every node is visited once
- Travel from node  $i$  to  $j$  cost  $c_{ij}$  (eg. travel time, distance, or cost)
  - Symmetric case  $c_{ij} = c_{ji} \forall i, j$ , suitable for inter-city transport
  - Asymmetric case  $c_{ij} \neq c_{ji}$  suitable for travel inside urban areas because of one-way streets

# Travel Salesman Problem: Bontour Case

Bontur is a pastry producer that was founded in Prague, Czech Republic, in the 19th century. The firm currently operates, in addition to four modern plants, a workshop in Gorazdova street where the founder started working. The workshop serves Prague and its surroundings. Every day at 6:30 AM, a fleet of vans carries the pastries from the workshop to several retail outlets (small shops, supermarkets and hotels). In particular, all outlets of the Vltava River district are usually served by a single vehicle. For the sake of simplicity,



# Travel Salesman Problem: Bontour Case Formulation

- **Input parameters:**
  - $S$  set of nodes (users) to be visited including the depot (pastry location 0),  $S = \{0,1,3,4,9,18,20,22\}$ ,
  - $c_{ij}$  cost of including the link from node  $i$  to  $j$  in the tour
- **Decision variables:** whether to include link  $i$  to  $j$  in the tour or,  $x_{ij}$  (binary: 1 if yes included, 0 otherwise),
- **Objective:** minimize total cost  $\sum_i \sum_j c_{ij} x_{ij}$ ,
- **Constraint 1:** Number of links going out of node  $i$  equals 1
$$\sum_j x_{ij} = 1, \forall i \text{ in } S,$$
- **Constraint 2:** Number of links going into node  $i$  equals 1
$$\sum_j x_{ji} = 1, \forall i \text{ in } S,$$



# Travel Salesman Problem: Bontour Case Solution

|                  | Cost (distance) from node I to j |           |           |           |           |           |           |           |
|------------------|----------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Input parameters | 0                                | 1         | 3         | 4         | 9         | 18        | 20        | 22        |
| 0                | <b>10</b>                        | 5.5       | 4.2       | 2.6       | 2.4       | 1.3       | 2.5       | 4.3       |
| 1                | 4.7                              | <b>10</b> | 3.7       | 2.1       | 5.1       | 6         | 7.2       | 9         |
| 3                | 4.2                              | 4.5       | <b>10</b> | 1.6       | 3.2       | 5.5       | 6.7       | 8.5       |
| 4                | 2.6                              | 2.9       | 1.6       | <b>10</b> | 3         | 3.9       | 5.1       | 6.9       |
| 9                | 3.8                              | 4.1       | 2.8       | 1.2       | <b>10</b> | 5.1       | 6.3       | 8.1       |
| 18               | 3.9                              | 7.4       | 6.1       | 4.5       | 3.3       | <b>10</b> | 1.2       | 3         |
| 20               | 3.5                              | 7         | 5.7       | 4.1       | 2.9       | 1.2       | <b>10</b> | 2.3       |
| 22               | 5.8                              | 9.3       | 8         | 6.4       | 5.2       | 3         | 2.3       | <b>10</b> |

|                     | x <sub>ij</sub> if link i to j is part of the tour |          |          |          |          |          |          |          | Auxilliary             |
|---------------------|----------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|------------------------|
| Decsion variables   | 0                                                  | 1        | 3        | 4        | 9        | 18       | 20       | 22       | Output links from node |
| 0                   | 0                                                  | 0        | 0        | 0        | 0        | <b>1</b> | 0        | 0        | 1                      |
| 1                   | 0                                                  | 0        | 0        | <b>1</b> | 0        | 0        | 0        | 0        | 1                      |
| 3                   | 0                                                  | 0        | 0        | 0        | <b>1</b> | 0        | 0        | 0        | 1                      |
| 4                   | 0                                                  | 0        | <b>1</b> | 0        | 0        | 0        | 0        | 0        | 1                      |
| 9                   | 0                                                  | <b>1</b> | 0        | 0        | 0        | 0        | 0        | 0        | 1                      |
| 18                  | <b>1</b>                                           | 0        | 0        | 0        | 0        | 0        | 0        | 0        | 1                      |
| 20                  | 0                                                  | 0        | 0        | 0        | 0        | 0        | 0        | <b>1</b> | 1                      |
| 22                  | 0                                                  | 0        | 0        | 0        | 0        | 0        | <b>1</b> | 0        | 1                      |
| Input links to node | 1                                                  | 1        | 1        | 1        | 1        | 1        | 1        | 1        |                        |

Solution includes 3 subtours (0 → 18 → 0, 1 → 4 → 3 → 9 → 1, 20 → 22 → 20) and does not satisfy the requirement of single tour

# Travel Salesman Problem: Bontour Case

## Subtours elimination

- Original solution includes 3 subtours ( $0 \rightarrow 18 \rightarrow 0$ ,  $1 \rightarrow 4 \rightarrow 3 \rightarrow 9 \rightarrow 1$ ,  $20 \rightarrow 22 \rightarrow 20$ ) and does not satisfy the requirement of single tour
- To eliminate subtours:
  - Select the shortest subtour that does not include depot (node 0)
  - Break the selected subtour by setting one of its links to zero
  - Resolve the TSP problem, if new solution includes subtours go to step 1, otherwise the solution is a feasible
- Applying this on Bontour Case, we find:

Lower bound on optimal

| Original solution (cost 20.8) |                      |    |    |    |   |
|-------------------------------|----------------------|----|----|----|---|
| Subtour 1                     | 0                    | 18 | 0  |    |   |
| Subtour 2                     | 1                    | 3  | 9  | 4  | 1 |
| Subtour 3                     |                      |    |    |    |   |
|                               | 20                   | 22 | 20 |    |   |
| Eliminate subtour 3           | $\times 20 \ 22 = 0$ |    |    |    |   |
| New solution                  |                      |    |    |    |   |
| Subtour 4                     | 0                    | 9  | 0  |    |   |
| Subtour 5                     | 1                    | 4  | 3  | 1  |   |
| Subtour 6                     | 18                   | 22 | 20 | 18 |   |

|              |                     |                      |   |    |    |    |   |   |   |   |   |  |  |  |  |  |  |  |  |
|--------------|---------------------|----------------------|---|----|----|----|---|---|---|---|---|--|--|--|--|--|--|--|--|
|              | Eliminate subtour 6 | $\times 20 \ 18 = 0$ |   |    |    |    |   |   |   |   |   |  |  |  |  |  |  |  |  |
| New solution | Subtour 7           |                      | 0 | 18 | 22 | 20 | 0 |   |   |   |   |  |  |  |  |  |  |  |  |
|              | Subtour 8           |                      | 1 | 4  | 3  | 9  | 1 |   |   |   |   |  |  |  |  |  |  |  |  |
|              | Eliminate subtour 8 | $\times 4 \ 3 = 0$   |   |    |    |    |   |   |   |   |   |  |  |  |  |  |  |  |  |
| New solution | Subtour 9           |                      | 0 | 18 | 22 | 20 | 0 |   |   |   |   |  |  |  |  |  |  |  |  |
|              | Subtour 10          |                      | 1 | 3  | 9  | 4  | 1 |   |   |   |   |  |  |  |  |  |  |  |  |
|              | Eliminate subtour 9 | $\times 4 \ 1 = 0$   |   |    |    |    |   |   |   |   |   |  |  |  |  |  |  |  |  |
| New Solution | Tour                |                      | 0 | 18 | 22 | 20 | 9 | 1 | 3 | 4 | 0 |  |  |  |  |  |  |  |  |
|              | Cost                | 21.5                 |   |    |    |    |   |   |   |   |   |  |  |  |  |  |  |  |  |

Solution obtained is not always optimal. Why?

Note the Quality of final solution can be checked by comparing its cost to lower bound

# Travel Salesman Problem: subtours elimination with additional constraints

- To eliminate subtours, label nodes visited in sequence in the tour, and enforce labels ordering to be in increasing way
- Node 0 is starting and terminal point in the tour so it does not satisfy the increasing ordering criteria (why?)
- Let  $u_j$  be node  $j$  ( $\neq 0$ ) label, a nonnegative number (**additional decision variable**)
- Linkage of labels of node  $i$  and  $j$  depends whether link  $i, j$  is part of tour or not:
  1. If link  $i, j$  is part of the tour then  $u_j \geq u_i + 1$
  2. Otherwise  $u_j, u_i$  are not connected (non binding)
- The if then condition can be written as

**Constraint 3:**  $u_j \geq u_i + 1 - M(1 - x_{ij})$ , for all  $i, j \in S - \{0\}$

1-5  $M$  is a big number, for example, equal to number of nodes

# Travel Salesman Problem: subtours elimination with additional constraints

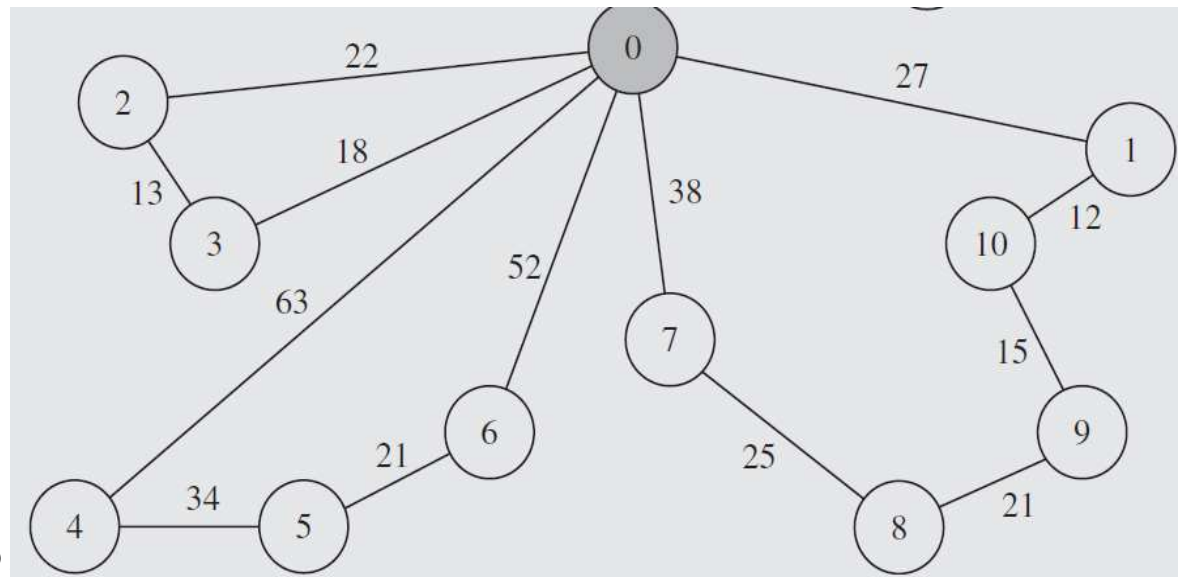
- Implement the labeling constraints by adding auxiliary variables (negative or 0)

$$u_i - u_j + 1 - M(1 - x_{ij}) \leq 0, \text{ for all } i, j \in S - \{0\}$$

- Solution using Excel Solver gives:
  - Optimal tour: 0,18,22,20,9,1,3,4,0
  - Optimal distance: 21.5
  - Another optimal tour: 0,18,22,20,9,3,4,1,0
  - Note, binary constraints are required to be added otherwise non binary  $x_{ij}$  are obtained

# Vehicle routing problem (VRP)

- Adding realistic constraints such as vehicle capacity, time window for delivery, limit on tour duration render the TSP problem infeasible (1 vehicle limit)
- In vehicle routing problem more than one vehicle start at the depot (node 0) and a set of nodes have to be visited
- Each vehicle serves a subset of nodes in mutual exclusive way



# VRP: Case Viola Oil Mill

- Italian oil mill Viola had to plan, starting from the production plant located in Venosa, the delivery of extra-virgin oil drums to its customers situated in 10 towns
- Customers are located in Spinazzola, Melfi, Rionero, Muro, Avigliano, Pietragalla, Acerenza, Tolve, Genzano, and Palazzo S. Gervasio.
- The amounts to be delivered per day to customers are given (measured in quintals 1 quintal = 1,000 KG ). Total is 83 quintals.
- A vehicle with a capacity of 35 quintals is used for the deliveries.
  - How many vehicles are at least needed?
- Distance matrix is given in Kilometers

Table 6.19 Order quantity (in q) received by Viola oil mill on March 14.

| Vertex | Town                 | Demand |
|--------|----------------------|--------|
| 1      | Spinazzola           | 9      |
| 2      | Melfi                | 8      |
| 3      | Rionero in Vulture   | 9      |
| 4      | Muro Lucano          | 10     |
| 5      | Avigliano            | 9      |
| 6      | Pietragalla          | 12     |
| 7      | Acerenza             | 8      |
| 8      | Tolve                | 6      |
| 9      | Genzano di Lucania   | 7      |
| 10     | Palazzo San Gervasio | 5      |

| Input parameters | Cost (distance in KM) from node I to j |            |       |         |      |           |             |          |       |         |         |
|------------------|----------------------------------------|------------|-------|---------|------|-----------|-------------|----------|-------|---------|---------|
|                  | Venosa                                 | Spinazzola | Melfi | Rionero | Muro | Avigliano | Pietragalla | Acerenza | Tolve | Genzano | Palazzo |
| Venosa           | 100                                    | 27         | 22    | 18      | 63   | 48        | 52          | 38       | 56    | 33      | 19      |
| Spinazzola       | 27                                     | 100        | 49    | 45      | 90   | 67        | 52          | 40       | 41    | 19      | 12      |
| Melfi            | 22                                     | 49         | 100   | 13      | 58   | 42        | 47          | 52       | 67    | 55      | 41      |
| Rionero          | 18                                     | 45         | 13    | 100     | 42   | 31        | 35          | 40       | 56    | 49      | 36      |
| Muro             | 63                                     | 90         | 58    | 42      | 100  | 34        | 62          | 75       | 67    | 85      | 81      |
| Avigliano        | 48                                     | 67         | 42    | 31      | 34   | 100       | 21          | 34       | 39    | 50      | 55      |
| Pietragalla      | 52                                     | 52         | 47    | 35      | 62   | 21        | 100         | 16       | 29    | 30      | 48      |
| Acerenza         | 38                                     | 40         | 52    | 40      | 75   | 34        | 16          | 100      | 25    | 15      | 32      |
| Tolve            | 56                                     | 41         | 67    | 56      | 67   | 39        | 29          | 25       | 100   | 21      | 38      |
| Genzano          | 33                                     | 19         | 55    | 49      | 85   | 50        | 30          | 15       | 21    | 100     | 15      |
| Palazzo          | 19                                     | 12         | 41    | 36      | 81   | 55        | 48          | 32       | 38    | 15      | 100     |



# VRP: Case Viola Oil Mill Formulation

- **Input parameters:**  $S$  set of customers to be visited including the plant (location 0),  $S = \{0,1,2,3,4,5,6,7,8,9,10\}$ .
  - $c_{ij}$  distance of the link from node  $i$  to  $j$  in a tour
  - $m$  number of vehicles to be used
  - $d_i$  demand of customer  $i$ ,  $C$  is vehicle limit
- **Decision variables:** whether to include link  $i$  to  $j$  in a tour or not,  $x_{ij}$  (binary: 1 if yes included, 0 otherwise),
- **Objective:** minimize total cost  $\sum_i \sum_j c_{ij} x_{ij}$ ,
- **Const 1:** Outgoing links of  $i$  ( $\neq 0$ ) equal 1:  $\sum_j x_{ij} = 1, i \in S - \{0\}$
- **Const 2:** Incoming links to  $i$  ( $\neq 0$ ) equals 1:  $\sum_j x_{ji} = 1, i \in S - \{0\}$
- **Const 3:** Outgoing links of 0 (plant) equals  $m$ :  $\sum_j x_{0j} = m$ ,
- **Const 4:** Incoming links to 0 (plant) equals  $m$ :  $\sum_j x_{j0} = m$ ,
- **Const 5:** Subtours elimination
- **Const 6:** vehicle capacity limitation



# VRP: Case Viola Oil Mill Formulation

- **Const 5:** Subtours elimination & **Const 6:** vehicle capacity limitation
  - Subtour elimination in TSP using labeling constraints can be used to satisfy vehicle capacity limit after modification
- Recall that the labels ( $u_i \geq 0$ ) in TSP where of increasing order in a tour and they satisfy

$$u_j \geq u_i + 1 - M(1 - x_{ij}), \forall i, j \in S - \{0\}$$

- In VRP with vehicle capacity limit, label  $u_i$  of a node  $i$  should be connected to the vehicle load as follows

$$u_j \geq u_i + d_j - C(1 - x_{ij}), \forall i, j \in S - \{0\}$$

$$u_i + d_i \leq C$$

- These last two constraints make sure no subtour exists and vehicle capacity is respected
- $u_i$  can be interpreted as an upper bound on the cumulative load in a tour up to node  $i$  (excluding node  $i$  load)

# VRP: Case Viola Oil Mill Solution

|    |                    | 0      | 1          | 2     | 3       | 4    | 5         | 6           | 7        | 8     | 9       | 10      |
|----|--------------------|--------|------------|-------|---------|------|-----------|-------------|----------|-------|---------|---------|
|    | Decision variables | Venosa | Spinazzola | Melfi | Rionero | Muro | Avigliano | Pietragalla | Acerenza | Tolve | Genzano | Palazzo |
| 0  | Venosa             | 0      | 0          | 1     | 0       | 1    | 0         | 0           | 1        | 0     | 0       | 0       |
| 1  | Spinazzola         | 0      | 0          | 0     | 0       | 0    | 0         | 0           | 0        | 0     | 0       | 1       |
| 2  | Melfi              | 0      | 0          | 0     | 1       | 0    | 0         | 0           | 0        | 0     | 0       | 0       |
| 3  | Rionero            | 1      | 0          | 0     | 0       | 0    | 0         | 0           | 0        | 0     | 0       | 0       |
| 4  | Muro               | 0      | 0          | 0     | 0       | 0    | 1         | 0           | 0        | 0     | 0       | 0       |
| 5  | Avigliano          | 0      | 0          | 0     | 0       | 0    | 0         | 1           | 0        | 0     | 0       | 0       |
| 6  | Pietragalla        | 1      | 0          | 0     | 0       | 0    | 0         | 0           | 0        | 0     | 0       | 0       |
| 7  | Acerenza           | 0      | 0          | 0     | 0       | 0    | 0         | 0           | 0        | 1     | 0       | 0       |
| 8  | Tolve              | 0      | 0          | 0     | 0       | 0    | 0         | 0           | 0        | 0     | 1       | 0       |
| 9  | Genzano            | 0      | 1          | 0     | 0       | 0    | 0         | 0           | 0        | 0     | 0       | 0       |
| 10 | Palazzo            | 1      | 0          | 0     | 0       | 0    | 0         | 0           | 0        | 0     | 0       | 0       |

Optimal Tours: (**1<sup>st</sup> vehicle**) Venosa, Melfi, Rionero, Venosa, (**2<sup>nd</sup> vehicle**) Venosa, Muro, Avigliano, Pietragalla, Venosa, (**3<sup>rd</sup> vehicle**) Venosa, Acerenza, Tolve, Genzano, Spinazzola, Palazzo, Venosa

$u_i$  (quintals)

| Spinazzola | Melfi | Rionero | Muro | Avigliano | Pietragalla | Acerenza | Tolve | Genzano | Palazzo |
|------------|-------|---------|------|-----------|-------------|----------|-------|---------|---------|
| 22.00      | 0.00  | 26.00   | 2.00 | 11.00     | 23.00       | 0.00     | 6.00  | 13.00   | 27.00   |

|                       |            |
|-----------------------|------------|
| <b>Objective (km)</b> | <b>357</b> |
|-----------------------|------------|

# VRP with additional constraints

- How to include limited shift time?
- How to include vehicles with different capacity?
- How to include in a tour that a pickup point should be visited before a delivery point?

# Key Terms (1 of 3)

- Accessorial service
- Barge
- Broker
- Common carrier
- Consignees
- Contract carrier
- Department of Transportation (DOT)
- Exempt carriers
- Freight forwarders
- Intermodal transportation
- Land bridge services

# Key Terms (2 of 3)

- Less-than-truckload (LTL)
- Line-haul
- Lock
- Parcel carriers
- Piggyback transportation
- Private carriers
- Rail gauge
- Shippers' associations
- Slurry systems
- Surface Transportation Board (STB)
- Twenty-foot equivalent unit (TEU)
- Terminals

# Key Terms (3 of 3 )

- Ton miles
- Transportation
- Truckload (TL)
- Unit load devices (ULDs)